

EVAN PENG

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EDUCATION

University of Alabama

Master of Arts in Mathematics

Bachelor of Science in Physics and Mathematics

Relevant Coursework: *Classical Mechanics (G)*, *Quantum Mechanics (G)*, *Electrodynamics*,
Intro Relativity, *Group Theory (G)*, *Ring Theory (G)*, *Real Analysis (G)*,
Measure Theory (G), *Topology (G)*

(Bold type indicates in progress courses, and (G) indicates graduate level courses)

Tuscaloosa, AL

Jan. 2026 – May 2028

Aug. 2024 – May 2028

RESEARCH INTERESTS

My research interests fall into theoretical physics. Particularly, I am interested in mathematical physics, but I want to explore research in more areas before I attend graduate school. I have found cosmology very interesting, but I want to gain exposure in high energy and condensed matter theory.

RESEARCH EXPERIENCE

Research Experience for Undergraduates

Louisiana State University

May 2026 – Present

Baton Rouge, LA

- **Project: Dynamical Systems Portrait of the Bounce in Loop Quantum Cosmology**

Advisor: Dr. Parampreet Singh

- * Analyzed a new model of Loop Quantum Cosmology proposed by Dr. Singh based on the polymer quantization of matter using both analytical and numerical methods
- * Developed background in Loop Quantum Gravity and Loop Quantum Cosmology, focusing on the new cosmological model proposed by Dr. Singh, including the ADM Formalism for GR, WDW canonical quantum gravity, Ashtekar variables, the holonomy-flux algebra, and polymer quantization
- * Rederived the equations of motion from the effective Hamiltonian for both standard LQC, and the new polymerized matter Hamiltonian
- * Performed Jacobian analysis on quasi-static points that compose a “loitering bounce” caused by the polymer matter, deriving a novel expression for a loitering timescale of the bounce
- * Programmed a system of equations solver and analyzer using SciPy for the given systems and analyzed the numerical evolution compared to the analytical predictions
- * Currently drafting a manuscript to be published

Undergraduate Research Assistant

University of Alabama

October 2024 – Present

Tuscaloosa, AL

- **Project: SingAura AI Voice Coach**

Advisor: Dr. Yi Lin

- * Hardware/Engineering Team Lead
- * Developed a custom biometric sensor array now under a **provisional patent** to optimally track respiratory activity for the purposes of classical singing
- * Led multiple undergraduate students to guide the progress and design of the device, along with the necessary data processing/analysis

TEACHING EXPERIENCE

Learning Assistant

University of Alabama

Jan. 2025 – Present

Tuscaloosa, AL

- **PH105:** General Physics with Calculus I
- **PH106:** General Physics with Calculus II

PRESENTATIONS

Summer Undergraduate Research Forum (SURF)

Louisiana State University

July 31, 2026

Baton Rouge, LA

- **Poster:** Dynamical Systems Portrait of the Bounce in Loop Quantum Cosmology
Authors: Evan Peng, Parampreet Singh

AWARDS AND HONORS

NSF Research Experience for Undergraduates	Summer 2026
President's List	Fall 2024 – Present
National Merit Scholar	2024 – Present

PROFESSIONAL MEMBERSHIPS

American Physical Society	May 2025 – Present
Society of Physics Students	May 2025 – Present
Pi Mu Epsilon Honor Society	May 2025 – Present

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, Matplotlib, PyTorch), C/C++, LaTeX, Git, R, MATLAB, Mathematica, Arduino
Foreign Language: Mandarin (Fluent Spoken)